Causal Forecast-Conditioned BS-RoFormer for Ocean Sound Separation

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***Abstract*—We test whether a background spectrum forecast from strictly preceding ocean audio helps separate the next 5-s acoustic event. FC-BSR injects the forecast into BS-RoFormer by concatenation, feature-wise linear modulation (FiLM), or cross-attention. A controlled 13-family study selects FiLM: at fixed 0-dB input SNR it improves SI-SDR improvement (SI-SDRi) by 0.692 dB (95% paired bootstrap interval $[0.542, 0.847]$). On 42 h of continuous hydrophone background, however, forecast conditioning is strength dependent: it exceeds the unconditioned model by 0.728 dB at -18 dB and 0.525 dB at -12 dB, then falls behind above approximately -4 dB. On 1,276 annotated real whale calls without isolated stems, FC-BSR slightly raises event-stem probability but does not reduce residual event probability or off-event change. Thus the forecast prior is useful in controlled and weak-target regimes, while its benefit on real unknown events remains unproven.**

***Index Terms*—ocean acoustics, source separation, soundscape forecasting, BS-RoFormer, weak-reference evaluation**

I. Introduction

Ocean backgrounds vary with shipping, weather, biology, and season [1]. A mixture-only separator must infer both an unexpected event and this changing background from the same observation. We instead ask whether the preceding soundscape can supply a causal prior:

$$S_{t-H:t-1} \xrightarrow{P_\psi} \widehat{S}_t, \qquad x_t = N_t + \alpha s_t \xrightarrow{F_\phi(\cdot|\widehat{S}_t)} (\hat{s}_t, \hat{N}_t), \quad (1)$$

where the separator never observes the current clean background N_t.

Time-domain separators [2], [3], attention models [4], [5], and band-split systems [6], [7] provide strong mixture-only backbones. Hybrid Transformers and MixIT broaden the architecture and supervision regimes [8], [9], but do not test a forecast of the upcoming background. FiLM, cross-attention, and cross-modal separation [10]–[12] motivate three interfaces. Forecasting baselines range from recurrent to linear and patch-based models [13]–[16].

Our contributions are: (1) FC-BSR,[1] with matched concatenation, FiLM, and cross-attention ablations; (2) a 13-family marine-aware source generator containing whale-like sweeps and vessel processes; (3) a 5-s real-background strength sweep with baseline, oracle, forecast, and shuffled conditions; and (4) a matched BS-RoFormer comparison on untouched annotated whale calls. The evidence is deliberately tiered so that synthetic ground truth is not conflated with semantic transfer.

[1]Code and reproduction instructions: https://github.com/D-eval/ocean_noise_pred_splite.

II. Method

A. Forecast and conditional separator

Historical one-third-octave levels (TOL) $S_{t-H:t-1} \in \mathbb{R}^{H\times F}$ are normalized using training data only. A residual GRU predicts

$$z_t = \mathrm{LN}(\mathrm{GRU}(S_{t-H:t-1})), \quad \widehat{S}_t = S_{t-1} + g_\psi(z_t). \quad (2)$$

Residual initialization represents persistence exactly; a learned predictor is retained only if validation MAE improves upon persistence [17].

BS-RoFormer maps the mixture STFT to band tokens X_k. At the same location before its axial Transformer, we compare

$$\begin{aligned} \text{Concat: } \widetilde{X}_k &= X_k + f([X_k; P_k(\widehat{S}_t)]), \\ \text{FiLM: } \widetilde{X}_k &= (1 + \tanh\gamma_k) \odot X_k + \beta_k, \\ \text{X-attn: } \widetilde{X} &= X + \mathrm{Attn}(X, E(\widehat{S}_t), E(\widehat{S}_t)). \end{aligned} \quad (3)$$

All residual branches begin at the identity. The loss combines waveform L_1 and multi-resolution complex-STFT losses; the TOL-surrogate study adds $0.2\|\widehat{S}_t - S_t\|_1$. Predictors are frozen before field-waveform separator training.

B. Marine-aware source distribution

The generator contains 13 families: harmonic, linear chirp, inharmonic, impulse train, frequency-modulated tone, band noise, whale upsweep, downsweep, arch, S-curve, pulse packet, vessel tonal, and vessel cavitation. Whale proxies use one to three localized events with variable duration, curvature, partials, taper, and flutter. Vessel-tonal and narrow-band cavitation components draw a 5–20 Hz low-frequency-oscillator amplitude envelope with probability 0.8; their broadband component is not modulated. Training SNR is uniform on $[-12, 12]$ dB. This distribution is intended to cover contour and modulation patterns, not to claim species identity.

We quantify spectral displacement with directional $D_{\mathrm{KL}}(p\|q)$ and finite symmetric Jensen–Shannon (JS) divergence [19]; spectral entropy and source–ocean overlap are reported separately. Larger KL/JS expands coverage but is not treated as semantic diversity.

C. Evidence ladder and protocol

Tier 1 uses NOAA–Navy SanctSound hourly TOL products [18]. MB03 deployments 01–02/03/04 are train/validation/test. Random-phase inversion creates controlled spectral surrogates, not recovered audio. Every waveform is 5 s; 4,000 examples per training deployment and 2,500 updates are used.

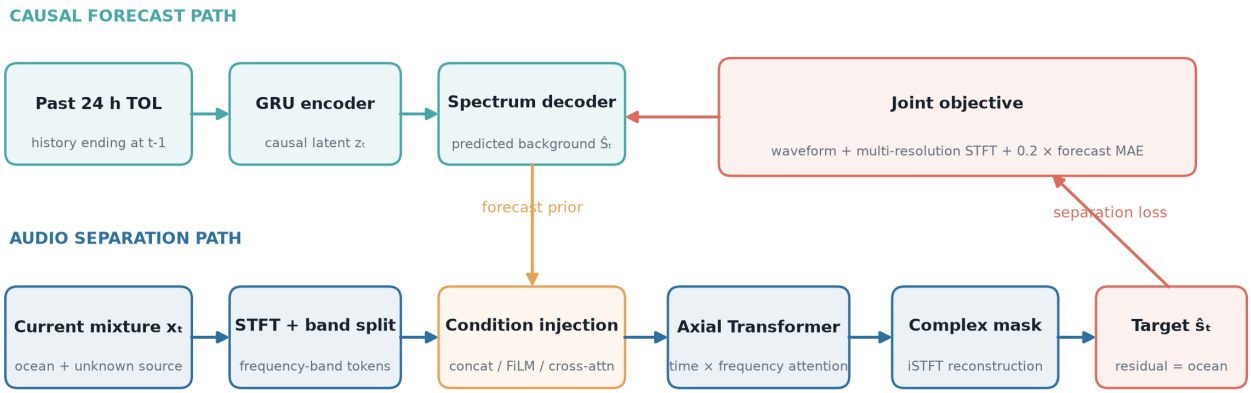


Fig. 1. FC-BSR. Strictly preceding soundscape features forecast the current background spectrum, which conditions a common BS-RoFormer backbone. Removing the blue path gives the parameter-matched unconditioned baseline.

Tier 2 uses 42 h of continuous 8-kHz hydrophone WAV (12/6/24 h for train/validation/test). The exact clean current waveform N_t is mixed with a synthetic target. The principal experiment uses 5-s mixtures and 2,500 updates, and evaluates 11 SNRs from -18 to $+12$ dB. A separate predictor-resolution diagnostic holds history duration at one hour and changes its step to 5 s, 30 s, 1 min, or 10 min; all later experiments return to 5-s waveforms.

Tier 3 uses day-disjoint 2-kHz DCLDE 2013 right-whale recordings [20]. Event annotations support detection [21], but isolated stems do not exist. Annotation-clean training clips receive the marine-aware proxies; 1,276 untouched real positive clips are evaluated. Both arms share samples, seeds, core initialization, 2,500-update budget, and validation-only checkpoint selection.

Complete-reference tiers report SI-SDR and SI-SDRi using scale-invariant projection [22], [23]. All uncertainty intervals are seeded paired bootstraps; continuous clips are block-bootstrapped by recording hour and real events by held-out day. Current test spectra are supervision or oracle controls only; causal histories end before t.

III. Experiments and Results

A. Conditioning selection and source diversity

Table I selects FiLM solely by validation performance. On 832 fixed-0-dB pairs (64 per family), it improves 4.850 to 5.541 dB SI-SDRi. Band-wise scale and bias can directly encode "expected energy here"; concat must compress doubled features, while cross-attention must learn alignment from a coarse condition under the same budget.

FiLM gains 0.869, 0.835, 0.919, 0.906, and 0.806 dB on the five whale proxy families. Vessel tonal is the hardest family: the conditioned absolute SI-SDRi is -8.295 dB, despite a 0.320-dB relative gain. Overall gain has weak correlation with JS ($r = -.057$) and entropy ($r = .122$), so increasing divergence broadens support but does not predict usefulness.

TABLE I
Matched 5-s conditioning ablation (dB).

Injection	Val. SI-SDRi	Test SI-SDRi	Paired Δ_0
None	4.864	4.059	0
Concatenation	5.343	4.432	+0.623
FiLM	**5.419**	**4.468**	**+0.692**
Cross-attention	5.128	4.261	+0.331

TABLE II
Predictor-resolution diagnostic on real background.

Step	Predictor	Cond.	Base	Gain [95% CI]
5 s	GRU	29.753	28.527	1.226 [1.082,1.381]
30 s	Persist.	29.630	28.534	1.095 [0.935,1.265]
1 min	Persist.	29.661	28.533	1.128 [0.971,1.298]
10 min	Persist.	29.711	28.535	1.176 [1.036,1.326]

Cond./Base/Gain are target SI-SDRi in dB.

B. Real background: time scale and target strength

The resolution diagnostic in Table II uses identical held-out examples. Only the 5-s predictor beats persistence on validation (MAE 1.366 versus 1.386 dB); the other scales select persistence. Although all conditioned separators beat their matched baselines in this diagnostic, the intervals overlap, so it does not establish a best time scale.

The harder fixed-5-s experiment compares BS-RoFormer, current-spectrum oracle, causal forecast, and time-shuffled condition on 1,152 paired test mixtures. Averaged across training SNRs, the baseline scores 23.470 dB and Forecast-FC-BSR 22.871 dB ($\Delta = -0.599$, CI $[-0.851, -0.346]$); even the oracle is lower. This exposes conditioning/capacity effects that the simpler diagnostic misses. Figure 2 gives the more informative strength sweep. Forecast conditioning helps weak targets by 0.728, 0.643, 0.525, 0.366, and 0.157 dB at $-18, -15, -12, -9, -6$ dB, respectively, then loses from -3 dB onward. The shuffled condition also helps the weakest

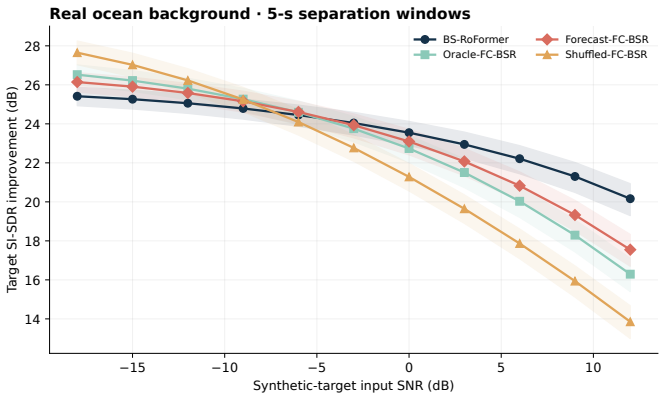


Fig. 2. Target-strength sweep on 5-s real ocean background. Lines show all four model conditions; shaded bands are paired 95% intervals.

TABLE III
MATCHED COMPARISON ON 1,276 REAL CALLS.

Method	$p(\hat{N})\downarrow$	$p(\hat{s})\uparrow$	Mono.↑	Off-event↓
BS-RoFormer	**.751**	.863	**.940**	**.279**
FC-BSR	.766	**.868**	.939	.288

targets but collapses as SNR rises; therefore weak-target gain alone cannot prove that temporal forecast accuracy, rather than regularization or extra capacity, is responsible.

C. Real events without separated ground truth

Following learned reference-free metrics [24], [25] and semantic audio embeddings [26], [27], we train, calibrate, and freeze a binary event classifier (test AUROC .9723, AP .9755, F1 .9275). Mixture consistency imposes $\hat{N} = x - \hat{s}$. We restore $\hat{s}$ at gains $\{-60, -30, -24, -18, -12, -6, 0\}$ dB and test whether event probability recovers monotonically. Identity and all-to-event controls reject trivial solutions; changes outside annotated intervals audit preservation.

The 12-by-5-s forecaster reduces validation MAE from 2.076 to 1.974 dB and is then frozen. On proxy validation, BS-RoFormer and FC-BSR score 11.125 and 11.088 dB SI-SDRi. Table III shows the untouched-call result. FC-BSR produces a slightly more event-like event stem ($+.0051$), but residual event probability is higher, monotonicity is unchanged, and off-event change increases. The day-block interval for residual probability difference $+.0145$ is $[-.0177, .0269]$. Hence this tier demonstrates separator transfer to real events, not an advantage from forecast conditioning.

IV. DISCUSSION AND CONCLUSION

The evidence supports a narrower claim than the initial hypothesis. FiLM is the best conditioning interface on controlled stems, and the prior helps weak targets in 5-s real backgrounds. It is not universally beneficial: the gain reverses for stronger targets, an oracle condition does not rescue the average, and no forecast advantage appears on untouched whale calls. Persistence also wins three of four predictor resolutions. These negative controls are central: they separate the existence of useful background information from a claim that the present architecture always exploits it.

The system is practical for inspection: all field-waveform variants run at RTF about .0028, and a local frontend displays forecasts and permits playback and download of both estimates. This does not establish field deployment. Future work should condition only under calibrated uncertainty, test multiple sites and seasons, and obtain isolated or stronger partial references for real events. Classifier-based restoration is evidence of semantic transfer, not a replacement for waveform fidelity.

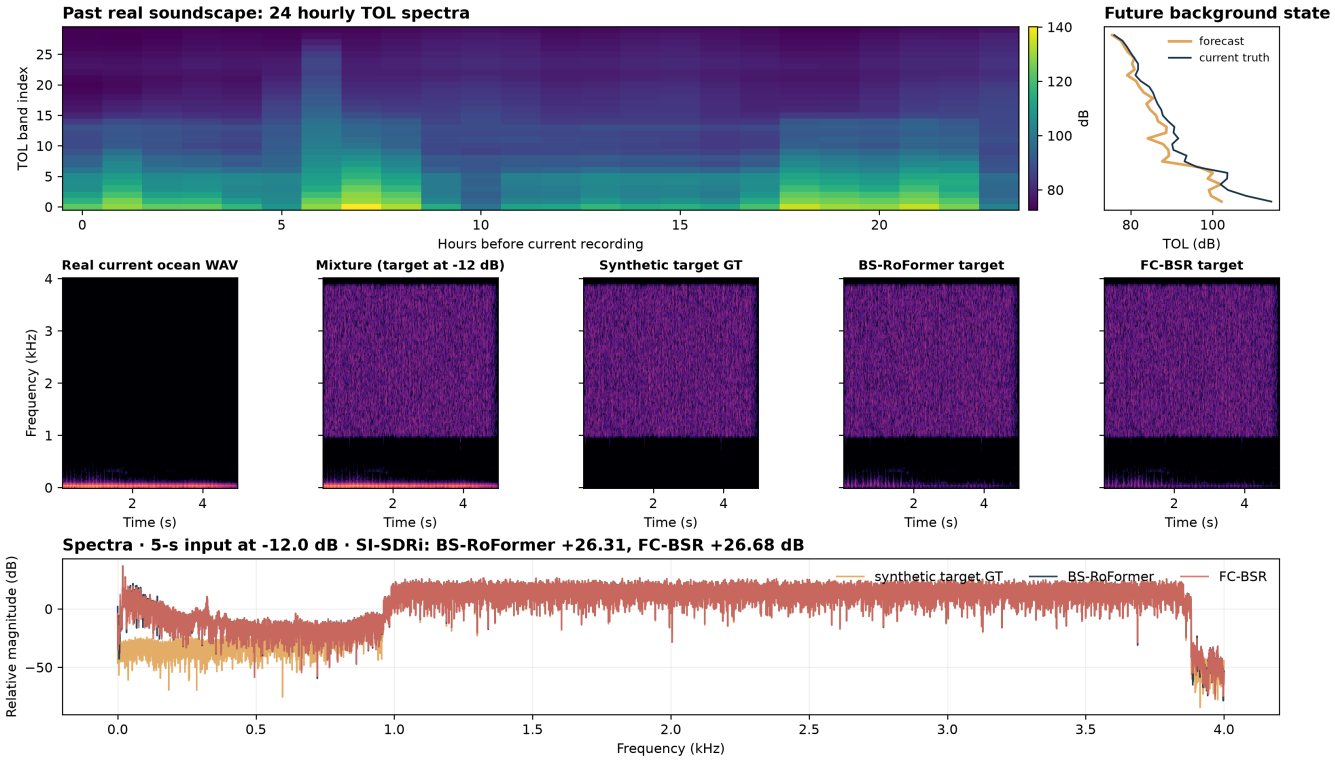


Fig. 3. A representative weak (−12 dB) 5-s target in real ocean audio. Unlike a visually dominant example, this view exposes the low-frequency background and compares BS-RoFormer and FC-BSR against the exact target.

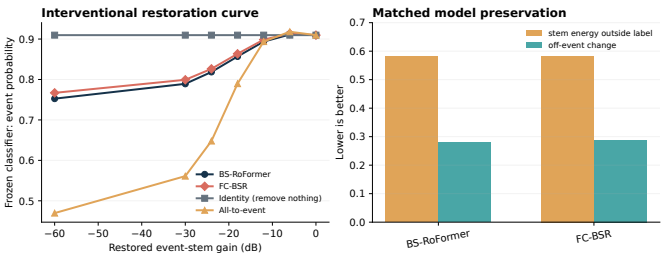


Fig. 4. Reference-free restoration curves and preservation audit. Both models perform nontrivial semantic separation, but FC-BSR is not uniformly better on real calls.

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